

FORM 2
THE PATENTS ACT, 1970
(39 of 1970)
&
The Patent Rules, 2003
COMPLETE SPECIFICATION
[See Section 10 and Rule 13]

1. TITLE OF THE INVENTION

“A TRANSITION METAL COMPLEX COMPOUND”

2. APPLICANT(S)

- a) **PI INDUSTRIES LTD.**
- b) an Indian company,
- c) of Udaisagar Road, Udaipur – 313001, Rajasthan, India

3. PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF THE INVENTION:

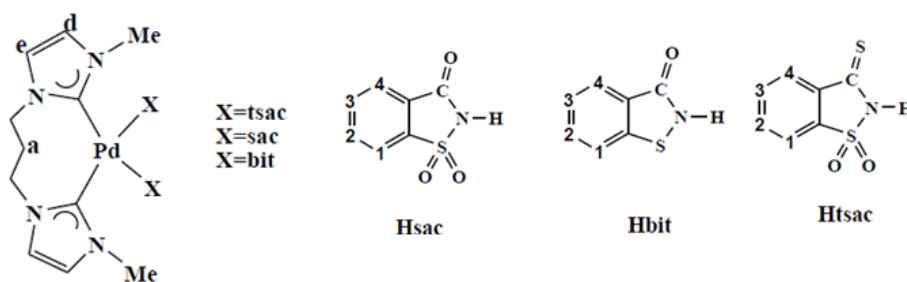
The present invention relates to a transition metal complex compound (**Z**), which is used as a catalyst for several chemical transformations.

BACKGROUND OF THE INVENTION:

The following discussion of the prior art is intended to present the invention in an appropriate technical context, and allows its significance to be properly appreciated. Unless clearly indicated to the contrary, reference to any prior art in this specification should not be construed as an expressed or implied admission that such art is widely known or forms part of common general knowledge in the field.

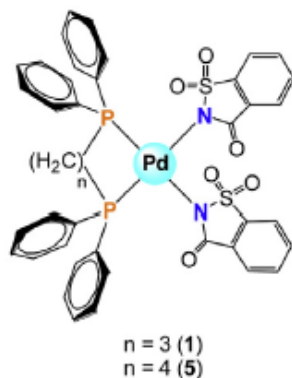
Transition metal complexes are extensively studied by scientists worldwide for the implication in various chemical transformations; hence there is a rapid development of newer process technologies relevant to industrial scale reactions for the production of organic compounds using transition metal complexes as catalyst. It is also observed that advanced research has been conducted globally in terms of formation of novel transition metal complexes as well as the use of known metal complexes as catalysts for critical chemical transformations.

Tikrit Journal of Pure Science, 22(2) 2017 describes synthesis and studies of Pd(II)-NHC complexes with thiosaccharinate, saccharinate or benzothiazolinate ligands as depicted below compound (**D**¹):



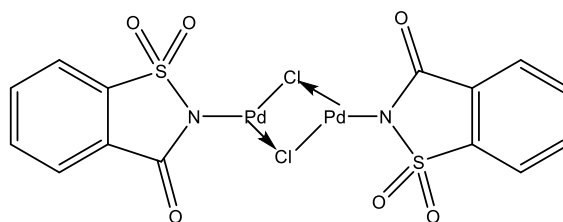
Compound (D**¹)**

European Journal of Medicinal Chemistry, 158, 534-547 (2018) describes Pd(II) and Pt(II) saccharinate complexes of bis(diphenylphosphino) propane or butane compound (**D**²) of formula:



Compound (D²)

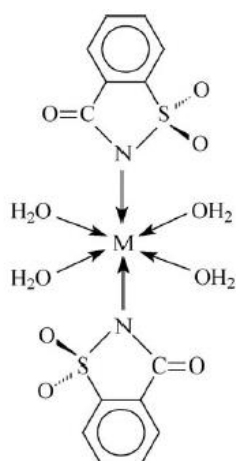
Journal of Molecular Catalysis A: Chemical, 269, 218–224 (2007) discloses sulfonamide and hydrazine-based palladium catalysts (as depicted below as compound (D³)), for effective use
 5 as catalysts for C–C coupling reactions. The article explicitly discloses the utility of palladium complex for the Suzuki coupling of 4-bromotoluene with phenylboronic acid.



Compound (D³)

U.S. Patent No. 7,999,112 (hereafter US'112) describes a reusable transition metal complex
 10 catalyst useful for the preparation of high quality 3,3'-diaminobenzidine and its analogues, comprises homocoupling of substituted 4-halo-2-nitroaniline in the presence of a transition metal complex catalyst.

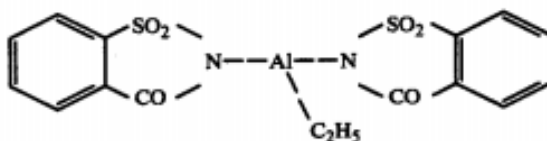
There are several other metal complexes reported in the art such as, Coordination Chemistry
 Reviews 250, 1980–1999 (2006) discloses certain saccharin based $[M(\text{sac})_2(\text{H}_2\text{O})_4] \cdot 2\text{H}_2\text{O}$
 15 complexes ($M = \text{V}, \text{Cr}, \text{Mn}, \text{Fe}, \text{Co}, \text{Ni}, \text{Cu}$ and Zn), also the article does not suggest the palladium (Pd)-saccharin complex. The compound as per Coordination Chemistry Reviews 2006 is depicted below as compound (D⁴);



Schematic drawing of the structure of the $[M(sac)_2(H_2O)_4] \cdot 2H_2O$ complexes ($M^{II} = V, Cr, Mn, Fe, Co, Ni, Cu$ and Zn).

Compound (D⁴)

US Patent No. 4,144,140 describes a method for the preparation of additives in organo-aluminum electrolyte media comprising a reaction of o-benzoic acid sulfonimide with aluminum triethyl to obtain di-(o-benzoylsulfonimide)-ethyl aluminum of the formula compound (D⁵) as depicted below;



Compound (D⁵)

In addition to afore discussed literature documents, there are a number of published patent and non-patent documents available that describe different metal complexes such as Bull. Chem. Soc. Jpn. 43, 3480 (1970); Oriental Journal of Chemistry, Vol. 35(1), 186-192 (2019); Inorganica Chimica Acta. 56, L31 (1981); Journal of Thermal Analysis and Calorimetry vol. 53(3), p843-854 (1998); Journal of Bangladesh Academy of Sciences vol. 37(2), p195-203 (2013); similarly Chinese patent application CN 103373972.

It is evident from the above discussion that several transition metal complexes are reported in the literature and being extensively used for coupling reactions. However, there is a continuous need to develop a stable and industrially viable transition metal complex.

Inventors of the present invention have developed a transition metal complex compound (**Z**) which is stable and can be effectively used as a catalyst in various chemical transformations, such as to prepare chemical intermediates, agrochemical compounds as well as pharmaceutical compounds.

OBJECTIVE OF THE INVENTION:

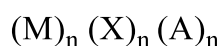
The main objective of the present invention is to provide a transition metal complex compound (**Z**).

Another objective of the present invention is to provide a palladium complex compound (**Z**¹).

- 5 Yet another objective of the present invention is to provide a palladium complex compound (**Z**²).

SUMMARY OF THE INVENTION:

In one aspect, the present invention relates to a transition metal complex compound (**Z**):



(Z)

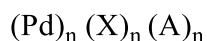
- 10 wherein **M** is a transition metal selected from Palladium (Pd), Platinum (Pt), Nickel (Ni), Copper (Cu), Cobalt (Co), Manganese (Mn), Vanadium (V), Zinc (Zn), Chromium (Cr), Cadmium (Cd), Rubidium (Rb), Lanthanum (La), Lead (Pb), Zirconium (Zr), Gold (Au), Mercury (Hg), Scandium (Sc) and Titanium (Ti),

X is a halogen selected from fluorine (F), chlorine (Cl), bromine (Br) and iodine (I),

- 15 n= 1-10,

A is an organic moiety selected from substituted or unsubstituted alkyl, alkenyl, alkynyl, carbocycle, heterocycle, aryl, heteroaryl, mono or bicyclic aromatic, mono or bicyclic heteroaromatic, saccharin, saccharin derivatives.

In one aspect, the present invention relates to a palladium complex compound (**Z**¹):



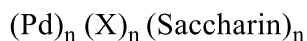
(Z¹)

- 20

wherein **X** is a halogen selected from fluorine (F), chlorine (Cl), bromine (Br) and iodine (I),
n= 1-10,

A is an organic moiety selected from substituted or unsubstituted alkyl, alkenyl, alkynyl, carbocycle, heterocycle, aryl, heteroaryl, mono or bicyclic aromatic, mono or bicyclic
25 heteroaromatic, saccharin, saccharin derivatives.

In another aspect, the present invention relates to a palladium complex compound (**Z**²):

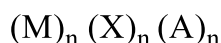


wherein **X** is a halogen selected from fluorine (F), chlorine (Cl), bromine (Br) and iodine (I),
n= 1-10.

In one aspect, the present invention provides a transition metal complex compound (**Z**) used as
5 a catalyst for hydrogenation reactions.

DETAILED DESCRIPTION OF THE INVENTION:

Accordingly, the present invention relates to a transition metal complex compound (**Z**):



wherein **M** is a transition metal selected from Palladium (Pd), Platinum (Pt), Nickel (Ni),
10 Copper (Cu), Cobalt (Co), Manganese (Mn), Vanadium (V), Zinc (Zn), Chromium (Cr),
Cadmium (Cd), Rubidium (Rb), Lanthanum (La), Lead (Pb), Zirconium (Zr), Gold (Au),
Mercury (Hg), Scandium (Sc) and Titanium (Ti),

X is a halogen selected from fluorine (F), chlorine (Cl), bromine (Br) and iodine (I),
n= 1-10,

15 **A** is an organic moiety selected from substituted or unsubstituted alkyl, alkenyl, alkynyl,
carbocycle, heterocycle, aryl, heteroaryl, mono or bicyclic aromatic, mono or bicyclic
heteroaromatic, saccharin, saccharin derivatives.

The meaning of various terms used in the description, and precisely relevant to define the scope
of organic moiety (**A**), shall be illustrated as follows:

20 Alkyl, alkenyl, alkynyl, carbocycle, heterocycle, aryl and heteroaryl groups, as defined herein,
are optionally substituted (e.g., "substituted" or "unsubstituted" alkyl, "substituted" or
"unsubstituted" alkenyl, "substituted" or "unsubstituted" alkynyl, "substituted" or
"unsubstituted" carbocyclyl, "substituted" or "unsubstituted" heterocyclyl, "substituted" or
"unsubstituted" aryl or "substituted" or "unsubstituted" heteroaryl group). In general, the term
25 "substituted", means that at least one hydrogen present on a group (e.g., a carbon or nitrogen
atom etc.) is replaced with a permissible substituent, e.g., a substituent which upon substitution
results in a stable compound, e.g., a compound which does not spontaneously undergo

transformation such as by rearrangement, cyclization, elimination, or other reaction under normal conditions (temperature, pressure, air etc.). Unless otherwise indicated, a "substituted" group has a substituent at one or more substitutable positions of the group, and when more than one position in any given structure is substituted, the substituent is either the same or different at each position.

The term "alkyl", used either alone or in compound words such as "alkylthio" or "haloalkyl" or -N(alkyl) or alkylcarbonylalkyl or alkylsulphonylamino or alkoxy or alkylsulfinyl includes straight-chain or branched C₁ to C₂₄ alkyl, preferably C₁ to C₁₅ alkyl, more preferably C₁ to C₁₀ alkyl, most preferably C₁ to C₆ alkyl. Representative examples of alkyl include methyl, ethyl, propyl, 1-methylethyl, butyl, 1-methylpropyl, 2-methylpropyl, 1,1-dimethylethyl, pentyl, 1-methylbutyl, 2-methylbutyl, 3-methylbutyl, 2,2-dimethylpropyl, 1-ethylpropyl, hexyl, 1,1-dimethylpropyl, 1,2-dimethylpropyl, 1-methylpentyl, 2-methylpentyl, 3-methylpentyl, 4-methylpentyl, 1,1-dimethylbutyl, 1,2-dimethylbutyl, 1,3-dimethylbutyl, 2,2-dimethylbutyl, 2,3-dimethylbutyl, 3,3-dimethylbutyl, 1-ethylbutyl, 2-ethylbutyl, 1,1,2-trimethylpropyl, 1,2,2-trimethylpropyl, 1-ethyl-1-methylpropyl and 1-ethyl-2-methylpropyl or the different isomers. If the alkyl is at the end of a composite substituent, as, for example, in cycloalkylalkyl, the part of the composite substituent at the start, for example the cycloalkyl, may be mono- or polysubstituted identically or differently and independently by alkyl. The same also applies to composite substituents in which other radicals, for example alkenyl, alkynyl, hydroxyl, halogen, carbonyl, carbonyloxy and the like, are at the end.

The term "alkenyl", used either alone or in compound words includes straight-chain or branched C₂ to C₂₄ alkenes, preferably C₂ to C₁₅ alkenes, more preferably C₂ to C₁₀ alkenes, most preferably C₂ to C₆ alkenes. Representative examples of alkenes include ethenyl, 1-propenyl, 2-propenyl, 1-methylethenyl, 1-butenyl, 2-butenyl, 3-butenyl, 1-methyl-1-propenyl, 2-methyl-1-propenyl, 1-methyl-2-propenyl, 2-methyl-2-propenyl, 1-pentenyl, 2-pentenyl, 3-pentenyl, 4-pentenyl, 1-methyl-1-butenyl, 2-methyl-1-butenyl, 3-methyl-1-butenyl, 1-methyl-2-butenyl, 2-methyl-2-butenyl, 3-methyl-2-butenyl, 1-methyl-3-butenyl, 2-methyl-3-butenyl, 3-methyl-3-butenyl, 1,1-dimethyl-2-propenyl, 1,2-dimethyl-1-propenyl, 1,2-dimethyl-2-propenyl, 1-ethyl-1-propenyl, 1-ethyl-2-propenyl, 1-hexenyl, 2-hexenyl, 3-hexenyl, 4-hexenyl, 5-hexenyl, 1-methyl-1-pentenyl, 2-methyl-1-pentenyl, 3-methyl-1-pentenyl, 4-methyl-1-pentenyl, 1-methyl-2-pentenyl, 2-methyl-2-pentenyl, 3-methyl-2-pentenyl, 4-methyl-2-pentenyl, 1-methyl-3-pentenyl, 2-methyl-3-pentenyl, 3-methyl-3-pentenyl, 4-methyl-3-pentenyl, 1-methyl-4-pentenyl, 2-methyl-4-pentenyl, 3-methyl-4-pentenyl, 4-methyl-4-

pentenyl, 1,1-dimethyl-2-butenyl, 1,1-dimethyl-3-butenyl, 1,2-dimethyl-1-butenyl, 1,2-dimethyl-2-butenyl, 1,2-dimethyl-3-butenyl, 1,3-dimethyl-1-butenyl, 1,3-dimethyl-2-butenyl, 1,3-dimethyl-3-butenyl, 2,2-dimethyl-3-butenyl, 2,3-dimethyl-1-butenyl, 2,3-dimethyl-2-butenyl, 2,3-dimethyl-3-butenyl, 3,3-dimethyl-1-butenyl, 3,3-dimethyl-2-butenyl, 1-ethyl-1-butenyl, 1-ethyl-2-butenyl, 1-ethyl-3-butenyl, 2-ethyl-1-butenyl, 2-ethyl-2-butenyl, 2-ethyl-3-butenyl, 1,1,2-trimethyl-2-propenyl, 1-ethyl-1-methyl-2-propenyl, 1-ethyl-2-methyl-1-propenyl and 1-ethyl-2-methyl-2-propenyl and the different isomers. "Alkenyl" also includes polyenes such as 1,2-propadienyl and 2,4-hexadienyl. This definition also applies to alkenyl as a part of a composite substituent, for example haloalkenyl and the like, unless defined specifically elsewhere.

The term "alkynyl", used either alone or in compound words includes straight-chain or branched C₂ to C₂₄ alkynes, preferably C₂ to C₁₅ alkynes, more preferably C₂ to C₁₀ alkynes, most preferably C₂ to C₆ alkynes. Non-limiting examples of alkynes include ethynyl, 1-propynyl, 2-propynyl, 1-butyne, 2-butyne, 3-butyne, 1-methyl-2-propynyl, 1-pentynyl, 2-pentynyl, 3-pentynyl, 4-pentynyl, 1-methyl-2-butyne, 1-methyl-3-butyne, 2-methyl-3-butyne, 3-methyl-1-butyne, 1,1-dimethyl-2-propynyl, 1-ethyl-2-propynyl, 1-hexynyl, 2-hexynyl, 3-hexynyl, 4-hexynyl, 5-hexynyl, 1-methyl-2-pentynyl, 1-methyl-3-pentynyl, 1-methyl-4-pentynyl, 2-methyl-3-pentynyl, 2-methyl-4-pentynyl, 3-methyl-1-pentynyl, 3-methyl-4-pentynyl, 4-methyl-1-pentynyl, 4-methyl-2-pentynyl, 1,1-dimethyl-2-butyne, 1,1-dimethyl-3-butyne, 1,2-dimethyl-3-butyne, 2,2-dimethyl-3-butyne, 3,3-dimethyl-1-butyne, 1-ethyl-2-butyne, 1-ethyl-3-butyne, 2-ethyl-3-butyne and 1-ethyl-1-methyl-2-propynyl and the different isomers. This definition also applies to alkynyl as a part of a composite substituent, for example haloalkynyl etc., unless specifically defined elsewhere. The term "alkynyl" can also include moieties comprised of multiple triple bonds such as 2,5-hexadiynyl.

The term "cycloalkyl" means alkyl closed to form a ring. Non-limiting examples include but are not limited to cyclopropyl, cyclopentyl and cyclohexyl. This definition also applies to cycloalkyl as a part of a composite substituent, for example cycloalkylalkyl etc., unless specifically defined elsewhere.

The term "cycloalkoxy", "cycloalkenyloxy" and the like are defined analogously. Non limiting examples of cycloalkoxy include cyclopropyloxy, cyclopentyloxy and cyclohexyloxy. This definition also applies to cycloalkoxy as a part of a composite substituent, for example cycloalkoxy alkyl etc., unless specifically defined elsewhere.

The term "halogen", either alone or in compound words such as "haloalkyl", includes fluorine, chlorine, bromine or iodine. Further, when used in compound words such as "haloalkyl", said alkyl may be partially or fully substituted with halogen atoms which may be the same or different. Non-limiting examples of "haloalkyl" include chloromethyl, bromomethyl, 5 dichloromethyl, trichloromethyl, fluoromethyl, difluoromethyl, trifluoromethyl, chlorofluoromethyl, dichlorofluoromethyl, chlorodifluoromethyl, 1-chloroethyl, 1-bromoethyl, 1-fluoroethyl, 2-fluoroethyl, 2,2-difluoroethyl, 2,2,2-trifluoroethyl, 2-chloro-2-fluoroethyl, 2-chloro-2,2-difluoroethyl, 2,2-dichloro-2-fluoroethyl, 2,2,2-trichloroethyl, pentafluoroethyl, 1,1-dichloro-2,2,2-trifluoroethyl, and 1,1,1-trifluoroprop-2-yl. This 10 definition also applies to haloalkyl as a part of a composite substituent, for example haloalkylaminoalkyl etc., unless specifically defined elsewhere.

The terms "haloalkenyl", "haloalkynyl" are defined analogously except that, instead of alkyl groups, alkenyl and alkynyl groups are present as a part of the substituent.

The term "hydroxy" means -OH , Amino means -NRR , wherein R can be H or any possible 15 substituent such as alkyl. Carbonyl means -C(O)- , carbonyloxy means -OC(O)- , sulfinyl means SO , sulfonyl means S(O)_2 .

The term "carbocycle" includes "aromatic carbocyclic ring system" and "nonaromatic carbocyclic ring system" or polycyclic or bicyclic (spiro, fused, bridged, nonfused) ring 20 compounds in which ring may be aromatic or non-aromatic (where aromatic indicates that the Huckel rule is satisfied and non-aromatic indicates that the Huckel rule is not satisfied).

The term "hetero" in connection with rings refers to a ring in which at least one ring atom is not carbon and which can contain 1 to 4 heteroatoms independently selected from the group consisting of nitrogen, oxygen and sulfur, provided that each ring contains no more than 4 nitrogens, no more than 2 oxygens and no more than 2 sulfurs.

25 The term "aromatic" indicates that the Huckel rule is satisfied and the term "non-aromatic" indicates that the Huckel rule is not satisfied.

The term "heterocycle" or "heterocyclic" or "heterocyclic ring system" includes "aromatic heterocycle" or "heteroaryl bicyclic ring system" and "nonaromatic heterocycle ring system" or polycyclic or bicyclic (spiro, fused, bridged, non-fused) ring compounds in which ring may 30 be aromatic or non-aromatic, wherein the heterocycle ring contains at least one heteroatom selected from N, O, S(O)_{0-2} , and or C ring member of the heterocycle may be replaced by C(=O) , C(=S) , $\text{C(=CR}^*\text{R}^*)$ and C=NR^* , * indicates integers.

The term “non-aromatic heterocycle” or “non-aromatic heterocyclic” means three- to fifteen-membered, preferably three- to twelve-membered, saturated or partially unsaturated heterocycle containing one to four heteroatoms from the group of oxygen, nitrogen and sulphur: mono, bi- or tricyclic heterocycles which contain, in addition to carbon ring members, one to three nitrogen atoms and/or one oxygen or sulphur atom or one or two oxygen and/or sulphur atoms; if the ring contains more than one oxygen atom, they are not directly adjacent; for example (but not limited to) oxetanyl, oxiranyl, aziridinyl, thietanyl, tetrahydrofuranyl, tetrahydrothienyl, pyrrolidinyl, isoxazolidinyl, isothiazolidinyl, pyrazolidinyl, oxazolidinyl, thiazolidinyl, imidazolidinyl, 1,2,4-oxadiazolidinyl, 1,2,4-thiadiazolidinyl, 1,2,4-triazolidin-1-yl, 1,2,4-triazolidin-3-yl, 1,2,3-triazolidinyl, 1,3,4-oxadiazolidinyl, 1,3,4-thiadiazolidinyl, 1,3,4-triazolidinyl, dihydrofuryl, dihydrothienyl, pyrrolinyl, isoxazolinyl, isothiazolinyl, dihydropyrazolyl, dihydrooxazolyl, dihydrothiazolyl, piperidinyl, pyrazynyl, morpholinyl, thiomorphlinyl, 1,3-dioxan-5-yl, tetrahydropyranyl, tetrahydrothienyl, hexahydropyridazinyl, hexahydropyrimidinyl, piperazinyl and cycloserines. This definition also applies to heterocyclyl as a part of a composite substituent, for example heterocyclylalkyl etc., unless specifically defined elsewhere. This definition also applies to heterocyclyl as a part of a composite substituent, for example heterocyclylalkyl etc., unless specifically defined elsewhere.

The term “heteroaryl” or “aromatic heterocyclic” means 5 or 6-membered, fully unsaturated monocyclic ring system containing one to four heteroatoms from the group of oxygen, nitrogen and sulphur; if the ring contains more than one oxygen atom, they are not directly adjacent; 5-membered heteroaryl containing one to four nitrogen atoms or one to three nitrogen atoms and one sulphur or oxygen atom: 5-membered heteroaryl groups which, in addition to carbon atoms, may contain one to four nitrogen atoms or one to three nitrogen atoms and one sulphur or oxygen atom as ring members, for example (but not limited thereto) furyl, thienyl, pyrrolyl, isoxazolyl, isothiazolyl, pyrazolyl, oxazolyl, thiazolyl, imidazolyl, 1,2,4-oxadiazolyl, 1,2,4-thiadiazolyl, 1,2,4-triazolyl, 1,3,4-oxadiazolyl, 1,3,4-thiadiazolyl, 1,3,4-triazolyl, tetrazolyl; nitrogen-bonded 5-membered heteroaryl containing one to four nitrogen atoms, or benzofused nitrogen-bonded 5-membered heteroaryl containing one to three nitrogen atoms: 5-membered heteroaryl groups which, in addition to carbon atoms, may contain one to four nitrogen atoms or one to three nitrogen atoms as ring members and in which two adjacent carbon ring members or one nitrogen and one adjacent carbon ring member may be bridged by a buta-1,3-diene-1,4-diyl group in which one or two carbon atoms may be replaced by nitrogen atoms, where these

rings are attached to the skeleton via one of the nitrogen ring members, for example (but not limited to) 1-pyrrolyl, 1-pyrazolyl, 1,2,4-triazol-1-yl, 1-imidazolyl, 1,2,3-triazol-1-yl and 1,3,4-triazol-1-yl.

- 5 6-membered heteroaryl which contains one to four nitrogen atoms: 6-membered heteroaryl groups which, in addition to carbon atoms, may contain, respectively, one to three and one to four nitrogen atoms as ring members, for example (but not limited thereto) pyridinyl, pyridazinyl, pyrimidinyl, pyrazinyl, 1,3,5-triazin-2-yl, 1,2,4-triazin-3-yl and 1,2,4,5-tetrazin-3-yl; benzofused 5-membered heteroaryl containing one to three nitrogen atoms or one nitrogen atom and one oxygen or sulphur atom: for example (but not limited to) indolyl, benzimidazolyl, 10 indazolyl, benzofuranyl, benzothiophenyl, benzothiazolyl, and benzoxazolyl; benzofused 6-membered heteroaryl which contains one to three nitrogen atoms: for example (but not limited to) quinolinyl, isoquinolinyl, quinoxalinyl, phthalazinyl, quinazolinyl and cinnolinyl. This definition also applies to heteroaryl as a part of a composite substituent, for example heteroarylalkyl etc., unless specifically defined elsewhere.
- 15 Bicyclic 5-6 heteroaryl systems with One bridgehead (Ring Junction) nitrogen atom containing one to three nitrogen atoms or one nitrogen atom and one oxygen or sulphur atom: for example (but not limited to) imidazo[1,2-a]pyridine, imidazo[1,2-a]pyrimidine, [1,2,4]triazolo[1,5-a]pyrimidine, [1,2,4]triazolo[1,5-b]pyridazine, [1,2,4]triazolo[1,5-a]pyrazine, [1,2,4]triazolo[1,5-a]pyridine, imidazo[1,2-c]pyrimidine, imidazo[1,2-b]pyridazine, 20 [1,2,4]triazolo[1,5-c]pyrimidine, 1-methyl-1H-indole, imidazo[1,2-a]pyrazine, pyrazolo[1,5-a]pyridine and [1,2,4]triazolo[4,3-a]pyridine.

- Non-limiting examples of fused 6-5-membered heteroaryl include Indolizinyl; pyrazolo[1,5-a]pyridinyl; imidazo[1,2-a]pyridinyl; pyrrolo[1,2-a]pyrimidinyl; pyrazolo[1,5-a]pyrimidinyl; imidazo[1,2-a]pyrimidinyl; pyrrolo[1,2-a]pyrazinyl; pyrazolo[1,5-a]pyrazinyl; imidazo[1,2- 25 a]pyrazinyl and the like.

This definition also applies to heteroaryl as a part of a composite substituent, for example heteroarylalkyl etc., unless specifically defined elsewhere.

- The “alkyl” also comprises “cyclic alkyl” or “cycloalkyl” which means alkyl closed to form a ring. Non limiting examples include but are not limited to cyclopropyl, cyclobutyl, cyclopentyl 30 and cyclohexyl. This definition also applies to cycloalkyl as a part of a composite substituent, for example cycloalkylalkyl etc., unless specifically defined elsewhere.

The “alkenyl” also comprises “cycloalkenyl” which means alkenyl closed to form a ring including monocyclic, partially unsaturated hydrocarbyl groups. Non limiting examples include but are not limited to cyclopentenyl and cyclohexenyl. This definition also applies to cycloalkenyl as a part of a composite substituent, for example cycloalkenylalkyl etc., unless specifically defined elsewhere.

The “alkynyl” also comprises “cycloalkynyl” which means alkynyl closed to form a ring including monocyclic, partially unsaturated groups. This definition also applies to cycloalkynyl as a part of a composite substituent, for example cycloalkynylalkyl etc., unless specifically defined elsewhere.

10 The term “bicyclic ring or ring system” denotes a ring system consisting of two or more common atom.

The term “aromatic” indicates that the Hueckel rule is satisfied and the term “non-aromatic” indicates that the Hueckel rule is not satisfied.

15 The terms “carbocycle” or “carbocyclic” or “carbocyclyl” include “aromatic carbocyclic ring system” and “nonaromatic carbocyclic ring system” or polycyclic or bicyclic (spiro, fused, bridged, nonfused) ring compounds in which the ring may be aromatic or non-aromatic (where aromatic indicates that the Huckel rule is satisfied and non-aromatic indicates that the Huckel rule is not satisfied).

20 Non limiting examples of non-aromatic carbocyclic ring system are cyclopropyl, cyclobutyl, cyclopentyl, norbornyl and the like. Non limiting examples of aromatic carbocyclic ring system are phenyl, naphthyl and the like.

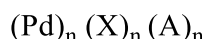
25 The term “aryl” as used herein is a group that contains any carbon-based aromatic group including, but not limited to phenyl, naphthalene, biphenyl, anthracene, and the like. The aryl group can be substituted or unsubstituted. In addition, the aryl group can be a single ring structure or comprise multiple ring structures that are either fused ring structures or attached via one or more bridging groups such as a carbon-carbon bond.

The term “aryl” also comprises “aralkyl” refers to aryl hydrocarbon radicals including an alkyl portion as defined above. Examples include benzyl, phenylethyl, and 6-naphthylhexyl. As used herein, the term “aralkenyl” refers to aryl hydrocarbon radicals including an alkenyl portion,

as defined above, and an aryl portion, as defined above. Examples include styryl, 3-(benzyl) prop-2-enyl, and 6-naphthylhex-2-enyl.

Any of the compounds according to the invention can also exist in one or more geometric isomer forms depending on the number of double bonds, chiral center or geometric rearrangement in the compound. The invention thus relates equally to all geometric isomers and to all possible mixtures, in all proportions. The geometric isomers can be separated according to general methods, which are known per se by a person ordinary skilled in the art.

In an embodiment, there is provided a palladium complex compound (**Z**¹):

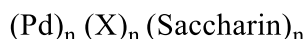


(**Z**¹)

10 wherein, **X** is a halogen selected from fluorine (F), chlorine (Cl), bromine (Br) and iodine (I), n= 1-10,

A is an organic moiety selected from substituted or unsubstituted alkyl, alkenyl, alkynyl, carbocycle, heterocycle, aryl, heteroaryl, mono or bicyclic aromatic, mono or bicyclic heteroaromatic, saccharin, saccharin derivatives.

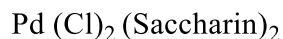
15 In yet another embodiment, the present invention relates to a palladium complex compound (**Z**²):



(**Z**²)

wherein, **X** is a halogen selected from fluorine (F), chlorine (Cl), bromine (Br) and iodine (I), n= 1-10,

20 In a specific embodiment, the transition metal complex is a palladium (Pd)-saccharin complex compound (**Z**³) represented by the formula:



(**Z**³)

In a specific embodiment, the process for the preparation of a palladium (Pd)-saccharin complex compound (**Z**³) is provided which comprises the steps of:

- 25 (i) suspending palladium chloride (PdCl₂) and lithium chloride in a solvent;
 (ii) stirring the reaction mixture of step (i);
 (iii) adding saccharin to the solution of step (ii) and continue stirring;

(iv) isolating the palladium (Pd)-saccharin complex compound (**Z**³).

The compound of the present invention as per the specific embodiment and its process described above is illustrated in the following **Scheme-1**,



Scheme-1

5 In an embodiment, a suitable solvent is selected from the group consisting of protic solvents, polar aprotic solvents and nonpolar solvents, including aromatic hydrocarbons, chlorinated hydrocarbons, ethers, aliphatic hydrocarbons, alcohols, esters, ketones, amides, water or mixtures thereof.

The solvent used in the step-(i) to step-(iv) of the above process (as depicted in the **Scheme-1**)
10 is selected from a halogenated solvent such as dichloromethane, 4-bromotoluene, diiodomethane, carbon tetrachloride, chlorobenzene and chloroform; alcoholic solvent such as methanol, ethanol, isopropanol, t-amyl alcohol, t-butyl alcohol and hexanol; an ether solvent such as tetrahydrofuran, cyclopentyl methyl ether, 2-methyltetrahydrofuran, diethyl ether and 1,4-dioxane; a ketone selected from methyl ethyl ketone, acetone; an aprotic solvent such as
15 acetonitrile, *N,N*-dimethyl formamide (DMF), *N,N*-dimethyl acetamide, dimethyl sulfoxide (DMSO) and *N*-methylpyrrolidone (NMP); an aromatic solvent such as toluene, xylene and benzene; acetone; water or a mixture thereof, and other polar as well as non-polar solvents.

The term ‘isolating the desired product’ referred to in the step (iv) of the above process (as depicted in the **Scheme-1**) corresponds to any of the steps involving biphasic separation,
20 separation of organic phase, filtration, evaporation of solvent, cooling, precipitation, washing and drying.

In an embodiment, the present invention provides a transition metal complex compound (**Z**) (as described herein) which is stable and can be effectively used as a catalyst in various chemical transformations, such as to prepare chemical intermediates, agrochemical compounds
25 as well as pharmaceutical compounds.

In a specific embodiment, the present invention provides a transition metal complex compound (**Z**) which is used as a catalyst for hydrogenation reactions.

In another embodiment, the present invention provides a transition metal complex (**Z**) which can be used as a catalyst in a heterogeneous catalysis reaction.

In one embodiment, the present invention provides a transition metal catalyst (Z) for reducing functional groups such as NO₂, CN, -C=N-, -C=C- or $\text{—C}\equiv\text{C—}$.

- 5 As used herein, the terms "comprises", "comprising", "includes", "including", "consisting" or any other variation thereof, are intended to cover a non-exclusive inclusion, subject to any limitation explicitly indicated. For example, a process or method that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process or method.
- 10 Also, the indefinite articles "a" and "an" preceding an element or component of the present invention are intended to be non-restrictive regarding the number of instances (i.e. occurrences) of the element or component. Therefore "a" or "an" should be read to include one or at least one, and the singular word form of the element or component also includes the plural unless the number is obviously meant to be singular.
- 15 The specification herein and the various features and advantageous details thereof are explained with reference to the non-limiting examples in the description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways in which the specification herein may be practiced and to further enable
- 20 those of skilled in the art to practice the specification herein. Accordingly, the examples should not be construed as limiting the scope of the specification herein.

Any discussion of documents, acts, materials, devices, articles and the like that has been included in this specification is solely for the purpose of providing a context for the disclosure. It is not to be taken as an admission that any or all of these matters form a part of the prior art

25 base or were common general knowledge in the field relevant to the disclosure as it existed anywhere before the priority date of this application.

The numerical values mentioned in the description and the foregoing claims though might form a critical part of the present invention of the present disclosure, any deviation from such

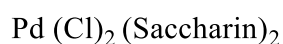
30 numerical values shall still fall within the scope of the present disclosure if that deviation

follows the same scientific principle as that of the present invention disclosed in the present disclosure.

The invention is further illustrated by the following examples which are provided to be exemplary of the invention, and do not limit the scope of the invention. While the present invention has been described in terms of its specific embodiments, certain modifications and equivalents will be apparent to those skilled in the art and are intended to be included within the scope of the present invention.

Examples:

Example 1: Preparation of palladium (Pd)-saccharin complex (Z³):



(Z³)

Palladium chloride (10 g, 0.056 M) and lithium chloride (5.64 g, 0.13 M) were added to methanol (126 mL). The reaction mixture was stirred for 2 hrs at 25 °C. A solution of saccharin (10.42 g, 0.057 M) and sodium acetate (6.94 g, 0.084 M) in methanol (100 g) was added to the above reaction mixture and stirred further for 8 hrs at 25 °C. The slurry was filtered to obtain desired compound (yield: 85%).

¹HNMR (400 MHz, DMSO- d₆): δ 7.56-7.76 (m, 8H);

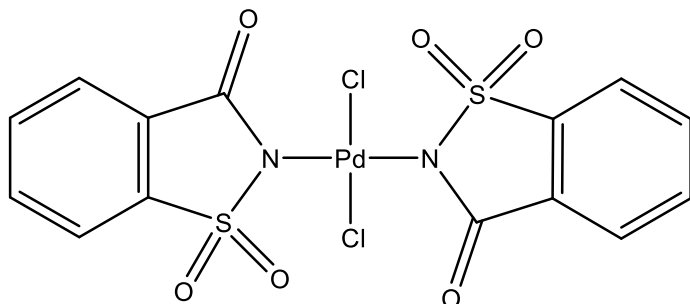
¹³C NMR (100 MHz, DMSO-d₆): 119.2, 120.0, 122.58, 123.09, 131.13, 131.67, 132.48, 132.80, 133.0, 134.73, 142.69, 145.28, 165.46, 168.06.

Elemental Analysis:

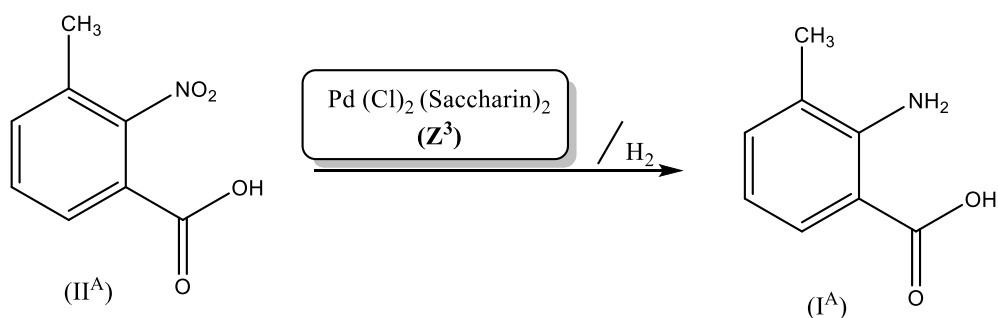
Elemental Analysis By CHNS Analyser								
					Instrument Name & Model		Thermoscientific / FlashSmart CHNS	
Molecular Weight(g)		541.62			EMP FORMULA		C ₁₄ H ₈ N ₂ O ₄ S ₂ Cl ₂ Pd	
Weight Of Sample (mg)					2.125	2.126		
Elements.	Amu	No. Of Elements	Theoretical	%	Observed %·R1	Observed %·R2	Average	Purity %
N	14.01	2	28.02	5.17	4.905	4.882	4.89	94.59
C	12.01	14	168.14	31.04	26.75	26.769	26.76	86.2
H	1.01	8	8.08	1.49	1.989	1.967	1.98	132.59
S	32.06	2	64.12	11.84	9.817	9.224	9.52	80.42
Cl	35.45	2	70.9	13.09				11.2

O	15.99	6	95.94	17.71				
Pd	106.42	1	106.42	19.65				16.13
Molecular Weight (g) calculated			541.62	100				

Structure:



Example 2: Preparation of 2-amino-3-methyl-benzoic acid (I^A):

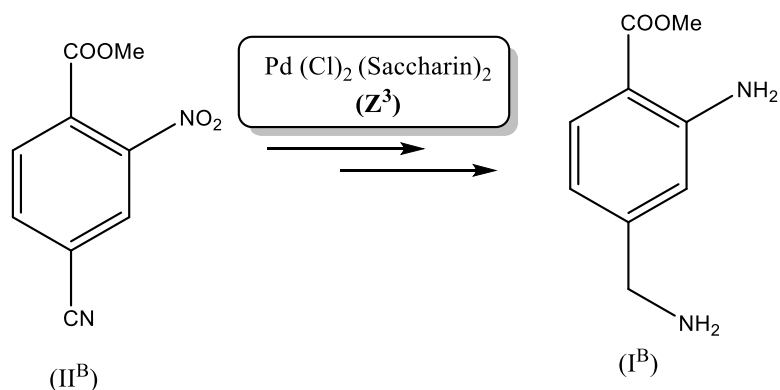


- 5 3-Methyl-2-nitrobenzoic acid (II^A) (8 g, 0.0055M) and the palladium (Pd)-saccharin complex (Z¹) (0.05 g) were added to an autoclave, the reaction mixture was stirred for 8 hrs at 25 °C under 20 kg/cm² of hydrogen gas pressure. The reaction mixture was filtered and the solvent was evaporated under reduced pressure. The residue was suspended in water (32 g) for 1 h to form slurry. The slurry was filtered and dried under reduced pressure to obtain the desired
- 10 compound I^A (yield: 97.3%).

¹H NMR (400 MHz, DMSO-*d*₆) δ 2.08 (s, 3H), δ 6.43-6.46 (t, 1H), δ 7.13-7.14 (d, 1H), δ 7.58-7.60 (d, 1H);

¹³C NMR (100 MHz, DMSO-*d*₆) 17.61, 109.48, 114.42, 123.08, 129.15, 134.52, 149.48, 170.19. MS m/z 152.03 [M+H]⁺.

- 15 **Example 3: Preparation of methyl 2-amino-4-(aminomethyl)benzoate (I^B):**



5 Methyl-4-cyano-2-nitrobenzoate (II^B) (2 g, 0.0097M) and the palladium (Pd)-saccharin complex (Z¹) (0.2 g) were added to an autoclave, the reaction mixture was stirred for 8 hrs at 25 °C under 20 kg/cm² of hydrogen gas pressure. The reaction mixture was filtered and the solvent was evaporated under reduced pressure to obtain the desired compound I^B (yield: 85%).

Example 4: Preparation of diethyl aminomalonate

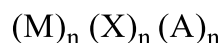
10 A solution of diethyl hydroxyiminomalonate (10.2 g, 0.054 mol) in methanol (40 g) was prepared. palladium (Pd)-saccharin complex (Z¹)(2 g) was added to the above solution and maintained 8 kg/cm² hydrogen gas in an autoclave for 5 h at 25 °C. After completion of the reaction, the catalyst was filtered, followed by evaporation of the solvent under reduced pressure to obtain diethyl aminomalonate (8.35 g, 90.5% yield).

¹H NMR (400 MHz, DMSO-*d*₆) δ 1.23 (t, 6H), δ 4.24-4.28 (m, 4H), δ 5.03 (s, 1H), δ 9.18 (bs, 2H);

¹³C NMR (100 MHz, DMSO-*d*₆), 13.81, 54.75, 62.84, 163.81. MS m/z 152.03 [M+H]⁺.

WE CLAIM:

1. A transition metal complex compound (**Z**):



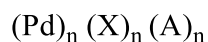
(Z)

wherein **M** is a transition metal selected from Palladium (Pd), Platinum (Pt), Nickel (Ni), Copper (Cu), Cobalt (Co), Manganese (Mn), Vanadium (V), Zinc (Zn), Chromium (Cr), Cadmium (Cd), Rubidium (Rb), Lanthanum (La), Lead (Pb), Zirconium (Zr), Gold (Au), Mercury (Hg), Scandium (Sc) and Titanium (Ti);

X is a halogen selected from fluorine (F), chlorine (Cl), bromine (Br) and iodine (I);
n= 1-10;

A is an organic moiety selected from substituted or unsubstituted alkyl, alkenyl, alkynyl, carbocycle, heterocycle, aryl, heteroaryl, mono or bicyclic aromatic, mono or bicyclic heteroaromatic, saccharin and saccharin derivatives.

2. The transition metal complex compound as claimed in claim 1, wherein M is palladium.
3. The transition metal complex compound as claimed in claim 2, wherein said transition metal complex compound represented as complex compound (**Z¹**):

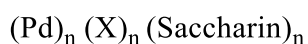


(Z¹)

wherein **X** is a halogen selected from fluorine (F), chlorine (Cl), bromine (Br) and iodine (I),
n= 1-10,

A is an organic moiety selected from substituted or unsubstituted alkyl, alkenyl, alkynyl, carbocycle, heterocycle, aryl, heteroaryl, mono or bicyclic aromatic, mono or bicyclic heteroaromatic, saccharin, saccharin derivatives.

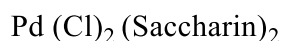
4. The transition metal complex compound as claimed in claim 1 or 3, wherein A is saccharin.
5. The transition metal complex compound as claimed in claim 4, wherein said transition metal complex compound represented as complex compound (**Z²**):



(Z²)

wherein **X** is a halogen selected from fluorine (F), chlorine (Cl), bromine (Br) and iodine (I),
and n= 1-10.

- 5 6. The transition metal complex compound as claimed in claim 1 or 5, wherein X is chlorine.
7. The transition metal complex compound as claimed in claim 6, wherein said transition metal complex compound represented as a palladium (Pd)-saccharin complex compound of formula (**Z**³):



(**Z**³)

- 10 8. The transition metal complex compound as claimed in claim 1, wherein said complex is used as a catalyst for hydrogenation reactions.
9. The transition metal complex compound as claimed in claim 1, wherein said complex as and when used for preparing a catalyst for use in a heterogeneous catalytic reaction.
- 15 10. A process for the preparation of a palladium (Pd)-saccharin complex compound (**Z**³) as claimed in claim 7, comprising the steps of:
- (i) suspending palladium chloride (PdCl₂) and lithium chloride in a suitable solvent;
- (ii) stirring the reaction mixture of step (i);
- (iii) adding saccharin to the solution of step (ii) and continue stirring;
- (iv) isolating the palladium (Pd)-saccharin complex compound (**Z**³).
- 20 11. The process as claimed in claim 10, wherein said suitable solvent is selected from the group consisting of protic solvents, polar aprotic solvents and nonpolar solvents, including aromatic hydrocarbons, chlorinated hydrocarbons, ethers, aliphatic hydrocarbons, alcohols, esters, ketones, amides, water or mixtures thereof.

25 Dated 26th day of July, 2022.



PRABHAKAR MANI PRATAP
of VUTTS & ASSOCIATES LLP
AGENT FOR THE APPLICANTS
IN/PA No. 1217

ABSTRACT

A TRANSITION METAL COMPLEX COMPOUND

The present invention relates to a transition metal complex (**Z**) (described herein), which is stable and can be effectively used as a catalyst in various chemical transformations such as to
5 prepare chemical intermediates, agrochemical compounds as well as pharmaceutical compounds.